

LangChain Framework Documentation

Official Reference Guide to LCEL Chains, Prompts, Loaders, and Runnables

1. LangChain Architecture & LCEL

LangChain is a framework for building LLM-powered applications. LangChain Expression Language (LCEL) uses the pipe operator (`|`) to compose prompts, chat models, tools, and output parsers into runnables.

```
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser

prompt = ChatPromptTemplate.from_messages([
    ("system", "You are a helpful assistant."),
    ("user", "{question}")
])

# LCEL Chain composition
chain = prompt | llm | StrOutputParser()
# response = chain.invoke({"question": "What is LangChain?"})
```

2. Vector Stores & Document Loaders

LangChain provides integrations for loading documents (PyPDFLoader, TextLoader) and chunking text via RecursiveCharacterTextSplitter prior to vector store indexing.

```
from langchain_text_splitters import RecursiveCharacterTextSplitter

splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200)
# docs = splitter.split_documents(raw_documents)
```